Probabilistic Machine Learning

Lecture 03: Keep the Peak, the Generative Route and Naive Bayes

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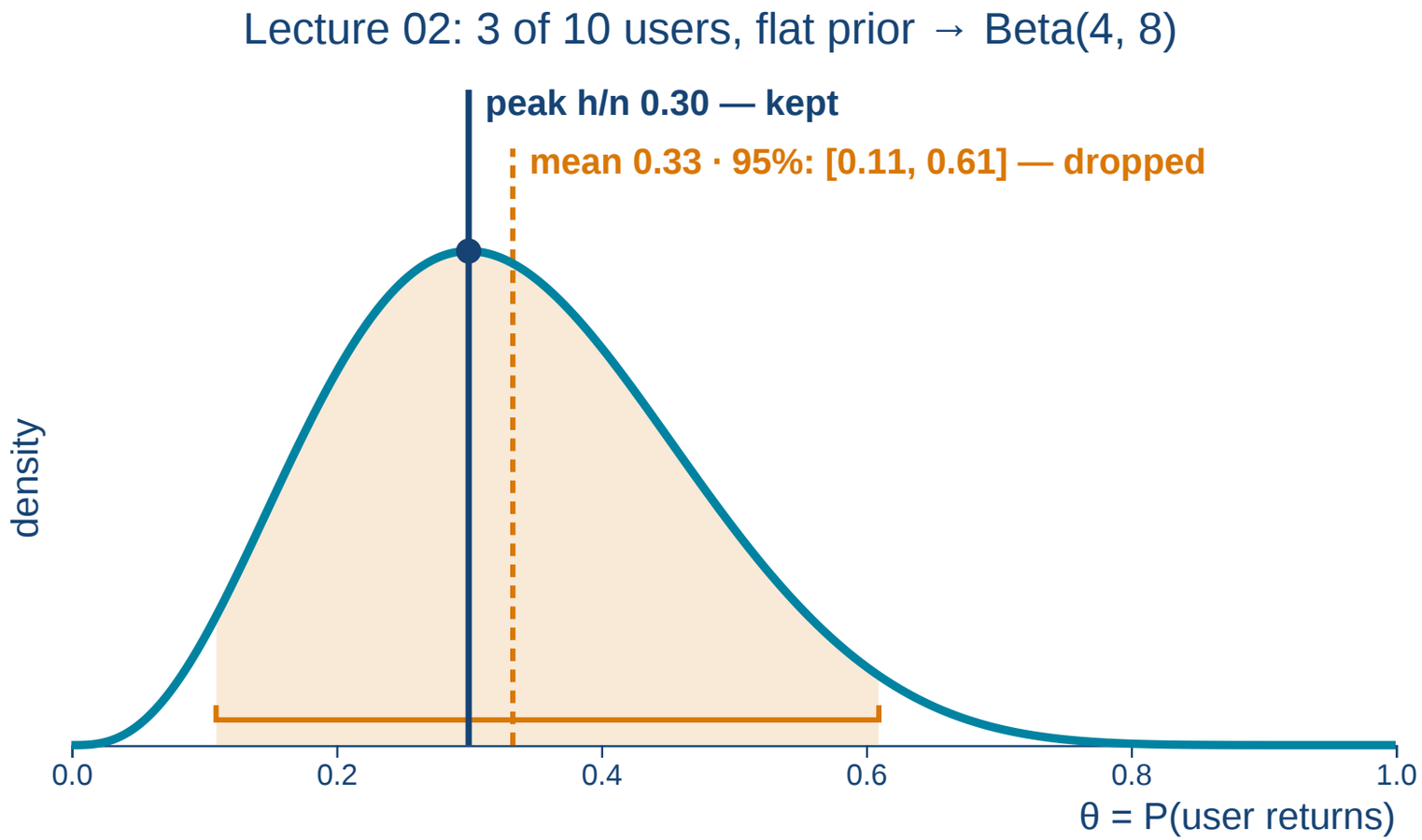
Speaker notes

One session. The first classifier of the course is built straight from probability, and two shortcuts are named on the way. Three modules: keep the peak (maximum likelihood and MAP on the Beta–Bernoulli coin of Lecture 02 — the tool that fitting needs, and the first thing a point estimate drops); the generative route and its explosion; and naive Bayes, read entirely through earlier material (fitting = MLE of a categorical, smoothing = the Beta posterior mean, double counting = over-confidence). Lecture 04 then shows that the linear and logistic regression the students know are the same story told for a conditional distribution — and sets logistic regression against naive Bayes.

2026-09-26 (course plan N2): conditional independence properly and the graphical-models introduction moved to the start of Lecture 05. This lecture defines conditional independence only as far as naive Bayes needs it. Fraud is the running example throughout; the shared Data Science units (bayes-decision-making, bayes-learning, naive-bayes) are replaced in PML by PML units and stay unchanged for Data Science.

2026-10-06: Lectures 03 and 04 swapped, so the course starts classification with a probabilistically motivated classifier. Keep-the-peak moved in here from the old Lecture 03; the comparison with logistic regression moved to Lecture 04.

Where We Left Off



One posterior, three readings:

- **peak** $0.30 = h/n$
- **mean** 0.33
- **95% interval** $[0.11, 0.61]$

Today keeps only the **peak** — and then asks what a classifier built on probability looks like.

Speaker notes

Pick up the thread of Lecture 02 deliberately; the last slides there promised this. Show the posterior the students computed last week, fully read: the peak (navy) is what today's methods keep, the mean and the interval (amber) are what they drop. Then say the uncomfortable sentence: every model they have fitted by least squares or cross-entropy took this shortcut — Lecture 04 returns to those. The next slide names the two shortcuts of the lecture.

Today: Keep the Peak, Assume It Apart

Lecture 01's table, rows 1 and 3 — and the first classifier built straight from probability:

shortcut	instead of the ideal	what is lost	where
Keep the peak	the most probable parameter — MLE or MAP	how sure we are about it	a coin · every count
Assume it apart	factorise the joint by independence	the dependences assumed away	naive Bayes

Keep the peak
on a coin

→

The generative route
a classifier from Bayes

→

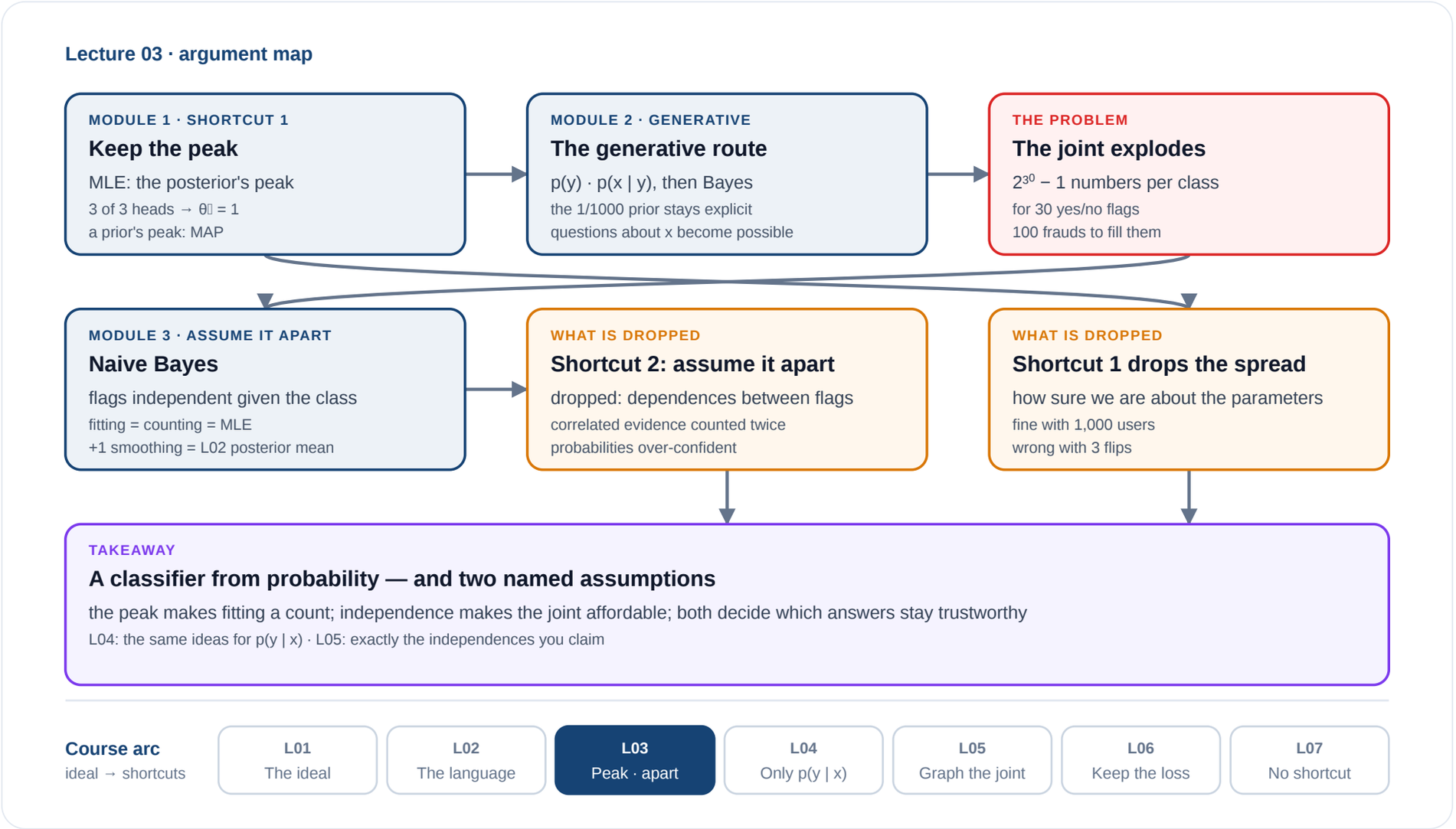
Naive Bayes
assume it apart

Speaker notes

The roadmap of the lecture in two forms: the two rows of the Lecture 01 table that today covers, and the three modules. Module 1 takes shortcut 1 where it can be drawn (the Beta posterior) and gives the tool, maximum likelihood, that fitting needs; Module 2 builds a classifier directly from Bayes' theorem and meets the explosion of the joint; Module 3 pays for it with the naive assumption and reads every step of naive Bayes as something already known (MLE, the Beta posterior mean, over-confidence). Lecture 04 then asks how the classifiers the students know from earlier courses — linear and logistic regression — look in this light.

Fraud is the running example: a base rate of 1 in 1,000 and a handful of yes/no red flags.

Argument Map



Speaker notes

The lecture on one page. Top row: keep the peak on a coin (and what it drops), the generative route, and the explosion of the joint it causes. Bottom row: naive Bayes as the payment, the shortcut named with what it drops, and the takeaway. Point at the strip: Lecture 03 is "keep the peak" and "assume it apart".

Keep the Peak

Guiding Question: Last week's posterior gave a peak, a mean and an interval. What do we lose when we keep only the peak — and why does almost everyone do it?

The Likelihood, Read Along the Other Axis

The **likelihood** is the probability of the observed data, read as a function of the **parameter** with the data held fixed: $L(\theta) = p(D \mid \theta)$. The **maximum likelihood estimate** is its peak, $\hat{\theta}_{\mathrm{ML}} = \arg\max_\theta p(D \mid \theta)$.

Lecture 02, 3 of 10 users

$$p(\theta \mid D) \propto \theta^3 (1-\theta)^7 \cdot 1$$

With a **flat prior** the posterior *is* the normalised likelihood — so its peak $h/n = 0.3$ is the maximum likelihood estimate.

Not a distribution over θ

$$\int_0^1 \theta^3 (1-\theta)^7 \mathrm{d}\theta = \frac{1}{1320}$$

The likelihood is a probability over **data**. Read along θ it need not integrate to 1 — only the posterior does.

Speaker notes

Start from the curve the students drew last week, not from a new definition. The likelihood is the one ingredient of the posterior that the data supplies; maximum likelihood keeps that ingredient, drops the prior (or takes it flat) and then keeps only the highest point. The integral makes the warning concrete: 1/1320 is B(4, 8), the Beta normaliser — the number the posterior divides by. The warning is the most common confusion of the course: "the likelihood of θ" is not "the probability of θ".

Deriving h/n Without a Posterior

Lecture 02's i.i.d. assumption lets the likelihood factorise over rows; the logarithm turns the product into a sum without moving the peak:

$$\ell(\theta) = \log \prod_{i=1}^{n} \theta^{y_i}(1-\theta)^{1-y_i} = h \log \theta + (n-h)\log(1-\theta)$$

$$\frac{d\ell}{d\theta} = \frac{h}{\theta} - \frac{n-h}{1-\theta} = 0 \quad \Longrightarrow \quad \hat{\theta}_{\mathrm{ML}} = \frac{h}{n}$$

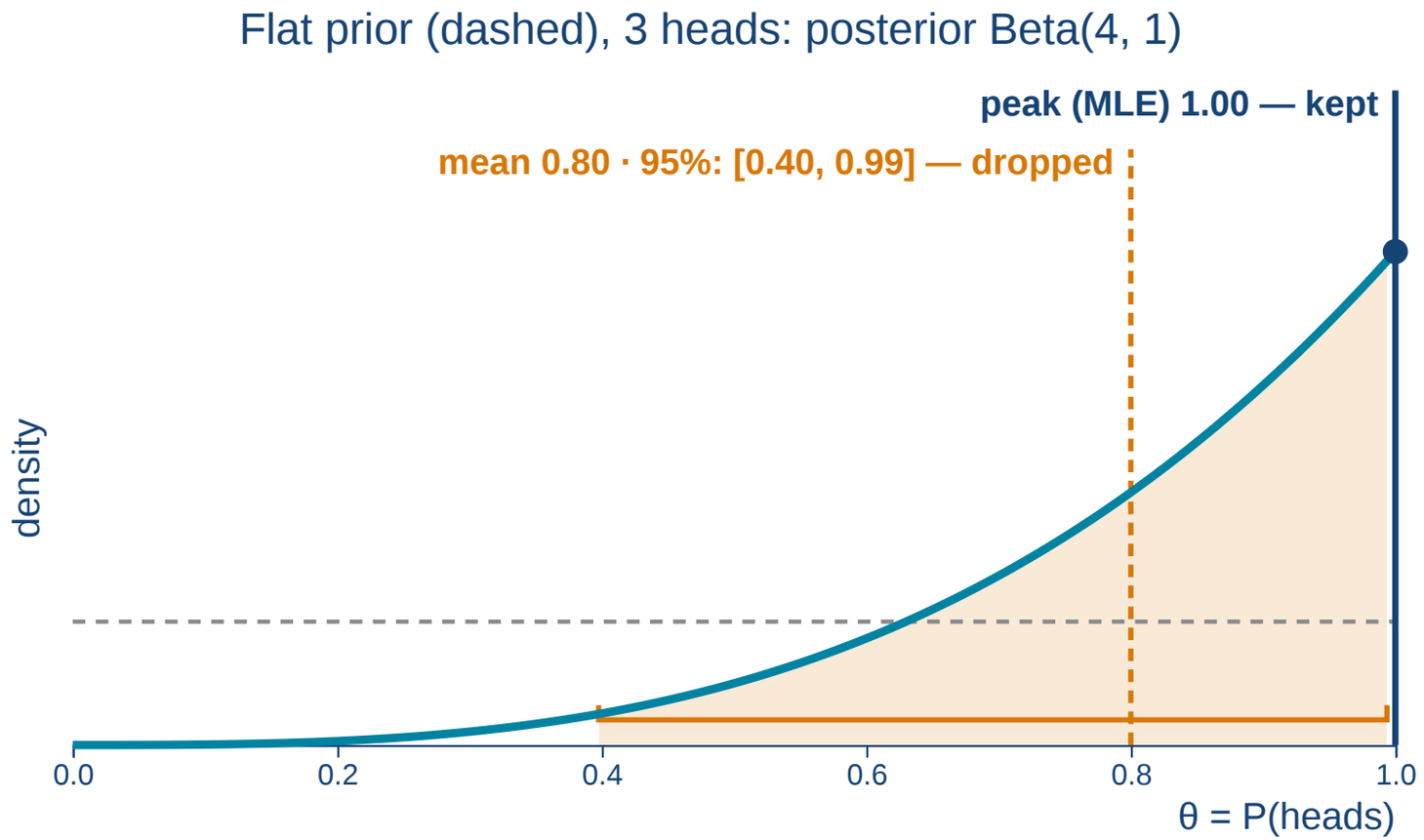
> Last week's peak, found by **setting a derivative to zero —** no prior, no normaliser, no integral. The recipe works unchanged for a million parameters, where nobody can draw the posterior.

Speaker notes

Math first: show the two lines and ask what they compute before saying it. The three reasons for the logarithm (same arg max, no underflow, sums differentiate term by term) can be said in one breath; students have met them in earlier ML courses. The insight box answers the second half of the guiding question: the peak is taken because it turns inference (an integral) into optimisation (a derivative). That is also why every method they know is fitted by gradient descent.

The Peak Can Lie

Three flips, **three heads**. Maximum likelihood: $\hat{\theta}_{\mathrm{ML}} = 3/3 = 1$.



Predict before the second click: where do the posterior mean and the 95% interval lie?

Speaker notes

Click 1 marks the peak — exactly 1: the estimate rules tails out forever. Stop and ask where the mean and the interval are. Click 2: mean 0.8, 95% interval [0.40, 0.99]. The posterior, with the same flat prior as last week, says "probably above a half, possibly well below 1"; the peak says "never tails". Nothing was wrong with the data or the model — only with keeping one number.

Numbers: Beta(4, 1) has density $4\theta^3$, CDF θ^4, so the 2.5% and 97.5% points are 0.025^(1/4) = 0.398 and 0.975^(1/4) = 0.994.

A Prior's Peak: MAP

Keep the prior, and still take only the peak — the **maximum a posteriori** estimate. With a $\mathrm{Beta}(a, b)$ prior, the same derivative gives

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} p(D \mid \theta) p(\theta) = \frac{h + a - 1}{n + a + b - 2}$$

prior	data	MLE h/n	MAP	posterior mean
$\mathrm{Beta}(1, 1)$ flat	3 of 3	1	1	0.80
$\mathrm{Beta}(2, 2)$	3 of 3	1	**0.80**	0.71
$\mathrm{Beta}(2, 2)$	8 of 10 (Lecture 02)	0.80	**0.75**	0.71

The prior shifts the peak as if $a - 1$ yeses and $b - 1$ noes had been seen before the data — a **flat** prior shifts nothing. MAP pulls 3 of 3 back to 0.8 — and still reports one number.

Speaker notes

The last row is the Lecture 02 mini-exercise (posterior Beta(10, 4)): students computed a MAP then without the name. Derivation for the curious: add $(a - 1) \log \theta + (b - 1) \log(1 - \theta)$ to $\ell(\theta)$ and set the derivative to zero. MAP is the bridge to Module 4, where a penalty on the weights turns out to be exactly this: a prior's peak. Note that the posterior mean $(a + h)/(a + b + n)$ is a different point — the shortcut chooses the mode, and for a skewed posterior the two differ.

Why Everyone Keeps the Peak

	the whole posterior	only the peak
needs	the normaliser $p(D)$ — an integral over θ	a derivative
in 1 dimension	a Beta table	h/n
in a million	an integral nobody can compute exactly	gradient descent
costs	computation	the **spread**

With **1,000 users** the posterior is $\mathrm{Beta}(301, 701)$: standard deviation 0.015. The peak is nearly everything there is to know.

Speaker notes

The trade in one table: the posterior asks for an integral, the peak for a derivative. With plenty of data the posterior is so narrow (Lecture 02: [0.27, 0.33]) that its peak is an excellent summary; with three flips it is not. The shortcut is not wrong, it is a bet that the spread does not matter — Module 2 shows the place where it matters even with plenty of data: far from the data.

Mini-Exercise

Two analysts both report $\hat{\theta} = 0.3$. One saw **3 of 10** users return, the other **300 of 1,000**.

What does the reported number hide — and which estimate would you bet on?

Speaker notes

One minute to read, one to think. Everything needed is on last week's slide; the arithmetic is not the point. Standard deviations 0.13 and 0.015 (ratio about 9).

Mini-Exercise

Two analysts both report $\hat{\theta} = 0.3$. One saw **3 of 10** users return, the other **300 of 1,000**.

What does the reported number hide — and which estimate would you bet on?

The **spread**. Lecture 02's posteriors: $[0.11, 0.61]$ after 10 users, $[0.27, 0.33]$ after 1,000 — the same peak, an interval nine times wider. Bet on the second; a point estimate cannot tell you to.

Takeaway

Shortcut 1 — keep the peak. Maximum likelihood (flat prior) and MAP (with a prior) keep the posterior's most probable point. It is cheap — a derivative instead of an integral — and harmless with plenty of data. It drops the **spread**: how sure we are about the parameter.

The Generative Route

Guiding Question: A classifier needs $P(\text{fraud} \mid x)$. Bayes' theorem builds it from how fraud *looks* — what would it take to model that, and why does nobody do it in full?

Two Routes to the Same Posterior

A labelled table of transactions supports **both** routes to $P(\text{fraud} \mid x)$, where x is the vector of red flags:

DISCRIMINATIVE — LECTURE 04

Model $p(y \mid x)$ directly

A score for each x, as logistic regression will give (Lecture 04). Nothing is said about how x arises.

GENERATIVE — TODAY

Model $p(y)$ and $p(x \mid y)$

How does a *fraudulent* transaction look, and how an honest one? Then convert with Bayes.

$$P(y \mid x) = \frac{p(x \mid y) \cdot P(y)}{\sum_{y'} p(x \mid y') P(y')}$$

Speaker notes

Math first on the second pass: the students know Bayes' theorem from L02; the only new thing is that the "hypothesis" is now a class and the "evidence" a whole attribute vector. The colour slots are those of L02 (posterior, likelihood, prior, normaliser).

Correct a misconception before it forms: the generative route is not forced on us by the data. A labelled table estimates p(y | x) just as directly — Lecture 03 did exactly that. The generative route is a *choice*: model each class separately, then convert.

Why Model How Fraud Looks?

A discriminative model — logistic regression, the next lecture — gives $P(\text{fraud} \mid x)$ directly. What can a model of $p(x \mid y)$ do that it cannot?

Speaker notes

Let the room answer before the click; students with an ML background usually name "missing values" and sometimes "the prior". The novelty point is the one they rarely name, and it is exactly what the discriminative route (Lecture 04) will drop.

Each of the three items returns in Lecture 04 as a concrete comparison. Here they only motivate paying the price on the next slide.

Why Model How Fraud Looks?

A discriminative model — logistic regression, the next lecture — gives $P(\text{fraud} \mid x)$ directly. What can a model of $p(x \mid y)$ do that it cannot?

- ✓ **Keep the prior explicit.** The 1 in 1,000 of Lecture 02 is a separate factor: a new market with 1 in 200 changes $P(y)$, not the whole model.
- ✓ **Handle a missing field.** An attribute nobody observed can be summed out of $p(x \mid y)$.
- ✓ **Notice the strange.** A transaction with tiny $p(x)$ under *both* classes is unlike anything seen — a score alone cannot say so.

> The price: $p(x \mid y)$ is a **joint distribution over all attributes**, one per class.

The Joint Explodes

30
yes/no red flags per transaction

→

$2^{30} - 1$
≈ 1.07 billion numbers for $p(x \mid \text{fraud})$ alone

→

100
frauds among 100,000 transactions to estimate them

Almost every cell of the table is **empty**. Its count is 0, so its maximum likelihood estimate is 0 — every fraud that looks slightly new becomes "impossible".

Speaker notes

The numbers: 2^30 − 1 = 1,073,741,823 free parameters per class (the cells of a joint over 30 binary variables, minus one because they sum to 1). With 100 frauds, at most 100 cells are non-empty. Connect to Module 1's three-heads-in-three-flips: MLE on too little data gives extreme answers; here it gives zeros almost everywhere.

The way out is not more data — no bank has a billion frauds. It is an assumption that makes the table small, which is Module 3.

Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

Speaker notes

Head arithmetic on purpose. If someone answers 16 or 17, the "minus one" (normalisation) is the point to make; it is the same count as the 2^n − 1 above. Keep the 15 on the board: Module 3 reduces it to 7.

Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

15. Each class needs a full table over $2^3 = 8$ flag patterns: 7 free numbers (they sum to 1). Two classes give 14, and $P(\text{fraud})$ adds one.

Takeaway

> The generative route models **how the data arises** — $p(y)$ and $p(x \mid y)$ — and converts with Bayes. It keeps the prior explicit and can answer questions about x itself, but it must pay for a **joint over all attributes**, which explodes.

Naive Bayes: Assume It Apart

Guiding Question: Its central assumption is almost always false, and it classifies well anyway. How can both be true?

Conditional Independence

Definition. A and B are **conditionally independent given** C, written $A \perp\!\!\!\perp B \mid C$, iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C) \quad \Longleftrightarrow \quad P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B tells us nothing more about A.**

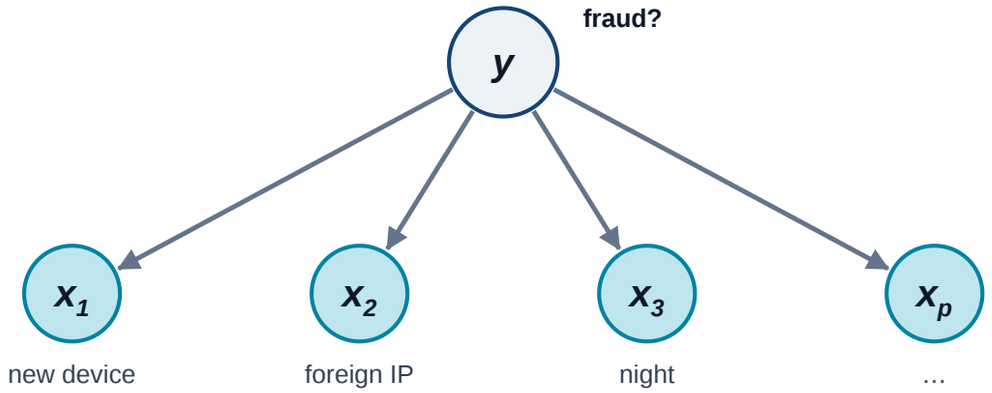
Among frauds only: does a new device make a foreign IP more likely?

Speaker notes

Math first, then the reading (P5). This is the definition naive Bayes rests on, so it comes before naive Bayes (the old order taught it a module later). The full treatment — a counted example, and the case where conditioning *creates* dependence — opens Lecture 05, where it becomes the language of graphs.

Leave the question open. Most students will say "yes, account take-overs bring both" — which is exactly the dependence naive Bayes will assume away, and the subject of "Counting Evidence Twice" at the end of the module.

The Naive Assumption



$p(y) \cdot p(x_1 \mid y) \cdot p(x_2 \mid y) \cdot \ldots \cdot p(x_p \mid y)$

Assume the attributes are **independent given the class**:

$$p(x_1, \ldots, x_p \mid y) = \prod_{j=1}^{p} p(x_j \mid y)$$

flags	full joint, per class	naive
3	7	3
30	1,073,741,823	30

Shortcut 3 — assume it apart. Dropped: every dependence *within* a class.

Speaker notes

The graph is a preview of Lecture 05, where a missing arrow will be read as an independence claim: here *no* arrow joins two attributes, so every pair is claimed independent given the class. Naive Bayes is the most extreme picture that is still a model.

The count: one Bernoulli per flag per class, p numbers instead of 2^p − 1. The mini-exercise of Module 1 (15 numbers for three flags) becomes 2 · 3 + 1 = 7.

Name the shortcut as in the Lecture 01 table and say what it drops. The model still keeps p(y) and p(x | y), so it is still generative; it just refuses to see how attributes move together.

Fitting Is Counting

$$\hat{P}(y) = \frac{n(y)}{n} \qquad \hat{P}(x_j = v \mid y) = \frac{n(x_j = v,\ y)}{n(y)}$$

Lecture 02

Each attribute, within one class, is a **categorical** variable — for a yes/no flag, a Bernoulli.

Module 1 today

Its maximum likelihood estimate is the **relative frequency** — the same derivation as h/n for the coin.

Naive Bayes is fitted in one pass over the table, with no optimiser — and every number it uses is a **peak**. Shortcut 1 is taken as well.

Speaker notes

Math first: show the two formulas and ask what principle produces them before revealing the cards verbally. The answer the course has prepared: the categorical distribution was introduced in L02 as "the distribution naive Bayes uses for every nominal attribute", and Module 1 derived that the MLE of a Bernoulli is h/n. Under the naive assumption the likelihood splits into one categorical per attribute and class, so each is maximised separately — by counting.

This is worth a sentence: the independence assumption is what makes the MLE a closed-form count. Logistic regression has no closed form.

The Classifier

$$P(y \mid x_1, \ldots, x_p) = \frac{P(y) \prod_j p(x_j \mid y)}{\sum_{y'} P(y') \prod_j p(x_j \mid y')}$$

For two classes, divide the two posteriors — the normaliser cancels:

$$\underbrace{\frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)}}_{\text{posterior odds}} = \underbrace{\frac{P(\text{fraud})}{P(\text{honest})}}_{\text{prior odds}} \cdot \prod_{j=1}^{p} \underbrace{\frac{p(x_j \mid \text{fraud})}{p(x_j \mid \text{honest})}}_{\text{likelihood ratio } \mathrm{LR}_j}$$

> Every observed attribute **multiplies the odds** by its likelihood ratio: above 1 it speaks for fraud, below 1 against.

Speaker notes

The first line is Bayes with the naive product, and the denominator is total probability — nothing new, only assembled. The odds form is what makes the method readable: each flag is one factor, and the order in which they arrive does not matter. Odds were introduced with Bayes in L02.

Three Red Flags

100,000 transactions, 100 of them fraud: prior odds $1 : 999$.

flag	among frauds	among honest	likelihood ratio
new device	60%	10%	6
foreign IP	70%	5%	14
night	50%	20%	2.5

A transaction shows all three flags. Is it more likely fraud than not?

Speaker notes

Let the room guess first; most will say "clearly fraud". The model here has exactly these three flags, all of them present, so only the three ratios enter. With a flag observed *absent* its factor would be (1 − 0.5)/(1 − 0.2) = 0.625 for night, for example — worth mentioning if someone asks.

Numbers: 0.6/0.1 = 6, 0.7/0.05 = 14, 0.5/0.2 = 2.5; 6 · 14 · 2.5 = 210; 210/(210 + 999) = 0.1737.

Three Red Flags

100,000 transactions, 100 of them fraud: prior odds $1 : 999$.

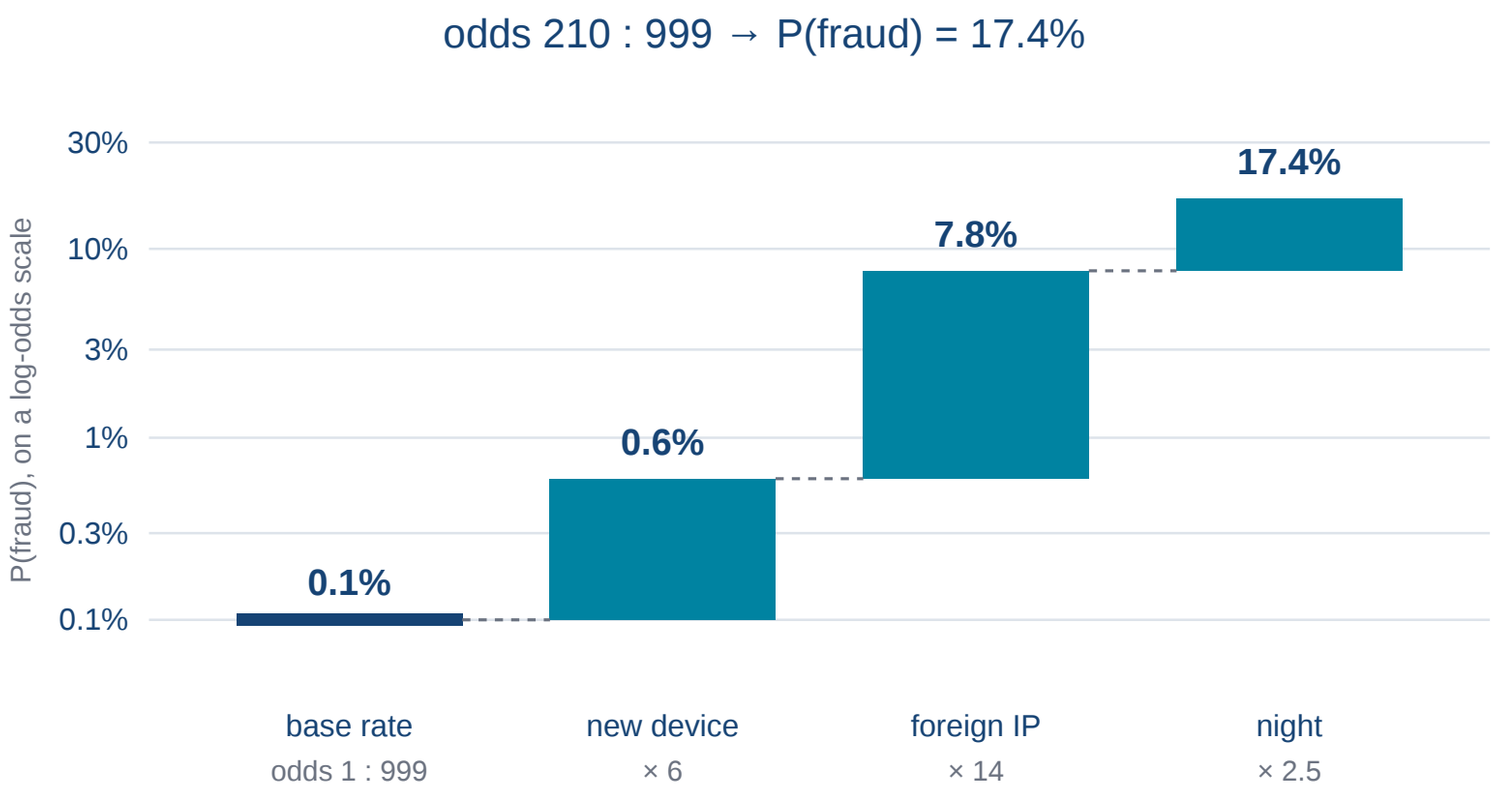
flag	among frauds	among honest	likelihood ratio
new device	60%	10%	6
foreign IP	70%	5%	14
night	50%	20%	2.5

A transaction shows all three flags. Is it more likely fraud than not?

$$\text{odds} = \frac{1}{999} \cdot 6 \cdot 14 \cdot 2.5 = \frac{210}{999} \quad \Longrightarrow \quad P(\text{fraud} \mid x) = \frac{210}{1{,}209} \approx 17\%$$

> Three red flags, each one strong — and still **mostly honest**. The base rate of Lecture 02, once more.

Evidence Adds Up



Predict before each click: which flag moves the probability most?

Speaker notes

Stepped: one click per flag. On a log-odds axis every likelihood ratio is a step of fixed height, independent of where we start: log 6, log 14, log 2.5. So the evidence *adds*. Stop after the foreign IP (7.8%) and ask what the night flag will do; students tend to overestimate it because "night" sounds suspicious, but 2.5 is the weakest ratio.

Plant the seed for Lecture 04: a sum of per-attribute contributions plus a starting value is a linear score — the form of logistic regression (Lecture 04).

One Zero Vetoes Everything

A new flag: **payment to a crypto exchange**. Among the 100 frauds it never occurred.

$$\hat{P}(\text{crypto} \mid \text{fraud}) = \frac{0}{100} = 0$$

A transaction with this flag gets posterior odds **exactly 0** — whatever the other thirty flags say. One unseen value vetoes all other evidence.

KEEP THE PEAK, FAILS

"Not seen in 100 frauds" became "impossible for a fraud" — the three heads in three flips of Module 1, in a single cell.

Speaker notes

The zero-frequency problem, in the course's language: an MLE on a small count is extreme. With 100 frauds, many plausible fraud patterns have count zero. Ask for a repair before the next slide — "add one" usually comes up. The next slide shows that "add one" is not a hack.

Smoothing Is a Prior

Laplace smoothing adds one pseudo-count to every value of attribute j (k_j values):

$$\hat{P}(x_j = v \mid y) = \frac{n(x_j = v,\ y) + 1}{n(y) + k_j}$$

For a yes/no flag ($k_j = 2$) this is $\frac{h+1}{n+2}$ — the **posterior mean** $\frac{a+h}{a+b+n}$ of Lecture 02 under the flat prior $\mathrm{Beta}(1, 1)$. For $k_j > 2$ values, the flat Dirichlet does the same.

count among 100 frauds	MLE (peak)	smoothed (mean)
0	0	$1/102 \approx 0.010$
60	0.60	$61/102 \approx 0.598$

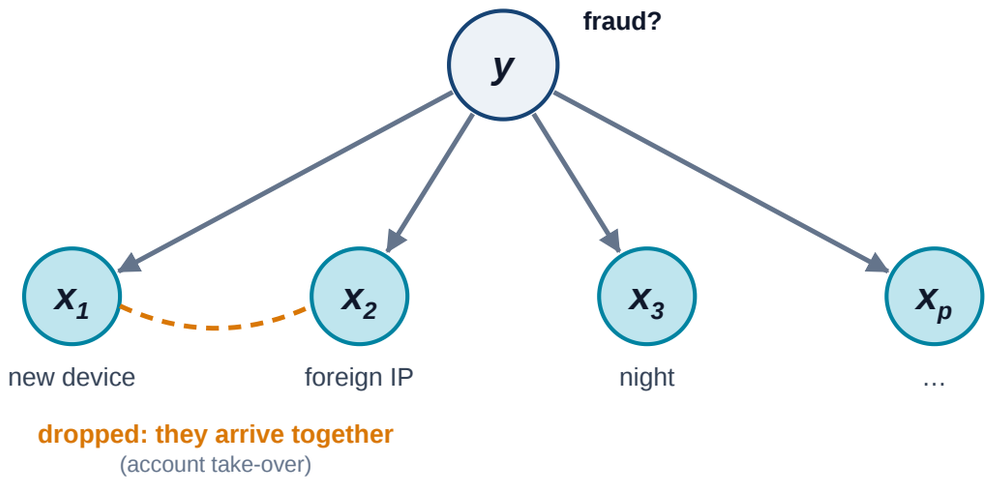
> Under the flat prior the peak of 0 of 100 is still 0. Smoothing quietly **leaves the peak for the mean** — exactly where the peak was wrong.

Speaker notes

This is the most satisfying tie of the lecture: a folklore trick of naive Bayes is the posterior mean the students computed in L02 (after 3 of 10 with a flat prior: mean 4/12 = 0.33). The Dirichlet is the Beta for more than two values; it needs only the name here.

The table shows why nobody notices the switch on well-populated cells (0.60 vs 0.598) and why it matters on empty ones. Larger pseudo-counts mean a stronger prior, as in L02's "near a half" analyst.

Counting Evidence Twice



In an account take-over, a new device and a foreign IP **arrive together**. Suppose both appear on 55% of frauds and 4% of honest transactions:

	ratio for the pair	$P(\text{fraud})$ **with night**
naive Bayes	$6 \cdot 14 = 84$	17.4%
the real pair	$0.55/0.04 \approx 14$	3.3%

Correlated flags are **counted twice**, so naive Bayes is **over-confident**. Its ranking of transactions often survives; its probabilities do not.

Speaker notes

Answer the question left open on the first slide. The dropped edge is drawn dashed: naive Bayes multiplies both ratios as if they were separate witnesses, though they largely report the same event. With the pair's own ratio 13.75, the posterior odds are 13.75 · 2.5/999 = 34.4/999, i.e. 3.3% instead of 17.4%.

The consequence connects to calibration (Lecture 04): naive Bayes scores push towards 0 and 1 and are not calibrated. That argmax decisions often survive anyway is the classic explanation of why naive Bayes classifies well (Domingos & Pazzani, 1997) — the answer to the module's guiding question.

Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

Speaker notes

One question, answerable qualitatively in seconds; the number is for checking (2940/3939 = 0.746). The bug is the extreme case of the previous slide: perfect correlation. Real feature sets contain near-copies all the time — the same information coded twice, derived features, one-hot groups.

Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

Up — from 17% to about 75%. The odds get a second factor 14: $2{,}940 : 999$. Naive Bayes cannot tell a copy from new evidence; a strongly correlated attribute is almost a copy.

Takeaway

Naive Bayes **assumes the joint apart** — attributes independent given the class — and **keeps the peak** of every count; smoothed, it keeps the Beta posterior *mean* instead. It is cheap, needs little data and ranks well. Its probabilities are over-confident wherever attributes move together.